

BENCHMARK: ACCURATE — fit consistent with published values

Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2458176.7242 +/- 0.0013 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.197 +/- 0.026                 |
| Transit depth (Rp/Rs)^2                           | 3.87 +/- 1.03 [%]               |
| Orbital Inclination (inc)                         | 79.58 +/- 2.0                   |
| Airmass coefficient 1 (a1)                        | 1.007 +/- 0.017                 |
| Airmass coefficient 2 (a2)                        | -0.0025 +/- 0.0065              |
| Scatter in the residuals of the lightcurve fit is | 1.67 %                          |
| Best Comparison Star                              | #3 - [147, 88]                  |
| Optimal Aperture                                  | 3.23                            |
| Optimal Annulus                                   | 7.55                            |
| Transit Duration (day)                            | 0.042 +/- 0.012                 |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.197 ± 0.026 vs 0.1594 (deviation 1.44σ) |
| Duration fit vs published | 0.042 d vs 0.0512 d (deviation 0.77σ)     |
| Mid-time O–C              | -3.7 min                                  |
| Depth SNR                 | 7.6                                       |
| Residual scatter          | 1.67 %                                    |

Light curve

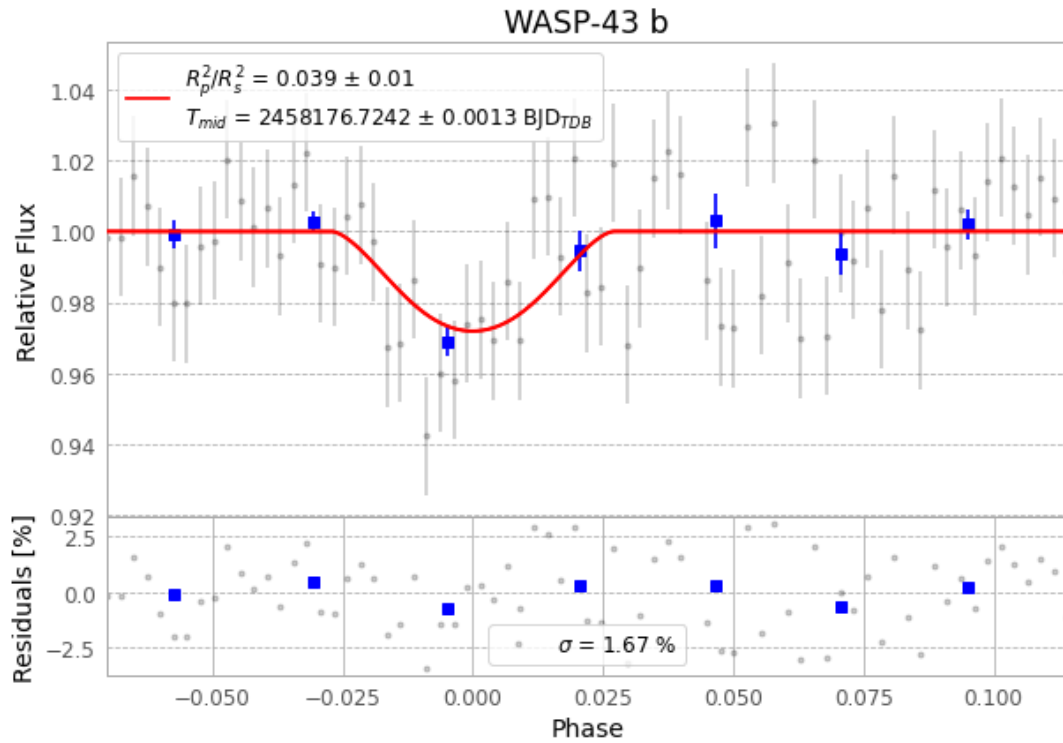


Figure 1 — FinalLightCurve\_WASP-43 b\_2018-02-26.png (SHA-256 22b2d84ab3525bc4...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [443, 212], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 568ea3bdd9ec814a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit b365494

## Data provenance

73 FITS frames (48 MB), WASP-43180227035112.FITS → WASP-43180227072712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180227.log (d0dd7a0ea556...), FinalLightCurve\_WASP-43 b\_2018-02-26.png (22b2d84ab352...), FinalLightCurve\_WASP-43 b\_2018-02-26.csv (7198b8c70ec5...)

## Compute resources

Wall time 110.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

---

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 10:55Z.